Common incidences in fifteen geometries of the Higman–Sims group

Let $G = \mathrm{HS}$ and $A = \mathrm{Aut}(G)$, of orders 44 352 000 and 88 704 000. We use their acquired faithful permutation models and the reviewed classification of coset geometries with specified parabolics $P_1, \ldots, P_r$. Source labels refer to the retained supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. The supplement attributes geometry 3.4 to Ivanov and Shpectorov; that attribution is retained.

Theorem 1. *The full groups are as follows. Here n is the least type size, d is the valency of the intrinsic graph used in the proof, and q is its full group order divided by $|G|$. Every type of size n is checked.*

Representative	*Source*	n	d	q	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
3	*2.4*	*5600*	*132*	*1*	G	A
4	*2.3*	*2200*	*105*	*4*	A	A
5	*2.2*	*4125*	*28*	*2*	A	A
7	*3.30*	*5600*	*132*	*1*	G	$G \times C_2$
8	*3.29*	*5600*	*132*	*1*	G	G
9	*3.28*	*5600*	*132*	*1*	G	G
13	*3.24*	*5600*	*495*	*1*	G	G
32	*3.4*	*3850*	*15*	*2*	A	A
33	*3.5*	*3850*	*15*	*2*	A	$A \times C_2$
42	*4.40*	*5600*	*132*	*1*	G	A
43	*4.36*	*5600*	*132*	*1*	G	G
45	*4.39*	*5600*	*132*	*1*	G	A
46	*4.41*	*5600*	*55*	*1*	G	A
73	*4.5*	*2200*	*105*	*4*	A	$A \times C_2$
78	*5.6*	*5600*	*132*	*1*	G	A

The correlation group in source diagram 4.41 is printed as $A \times C_2$; the exact calculation below gives A. The other group assertions in this table agree with the source, with an omitted correlation entry understood according to the supplement's stated convention.

Counting pairs from one point. Choose an anchor type a of size n, base point b, and stabilizer P_a. Let O_k be its orbits on points, with $d_k = |O_k|$. For another type j, its base object has point shadow $S_j = b^{P_j}$. For a non-diagonal orbit, let e_{jk} count ordered pairs in S_j belonging to the G-orbit of (b, y), where $y \in O_k$. Then

$$e_{jk} = |S_j|\,|S_j \cap O_k|, \qquad m_{jk} = \frac{[G : P_j]|S_j|\,|S_j \cap O_k|}{n d_k}. \tag{1}$$

To prove the first equality, take any $x \in S_j$. Some element of P_j sends b to x. It preserves S_j and the specified G-orbit on pairs, so the number of possible second coordinates at x equals that at b. Summing this constant over S_j proves the assertion. For the second, count triples consisting

of a type-j object and a pair in its shadow. There are $[G : P_j]e_{jk}$ triples and nd_k pairs, and transitivity makes the number of objects per pair constant. This counts distinct objects even when their shadows coincide. The type-a convention is equality within that type; its off-diagonal entries are zero.

For each pair form the sorted multiset of $([G : P_j], m_{jk})$ over all types. A correlation preserves this signature, including multiplicities. Partition the distinct pairs by signature and order the signatures lexicographically. For the valency d in the table, choose the first relation with that valency. This rule is intrinsic and is compatible with passage to another anchor of the same size. It does not require the valency to determine a unique relation: in representative 73 there are two relations of valency 105. Every selected relation is verified to be symmetric and irreflexive.

Exact point coordinates and faithful restriction. For each anchor a literal ordered list of generator images defines a bijective homomorphism $\phi : G \longrightarrow E \leq \mathrm{Sym}(n)$. The saved elements $t_x \in G$ satisfy $b^{\phi(t_x)} = x$, and the stabilizer of b in E is exactly $\phi(P_a)$. These checks prove the identification of the right coset $P_a g$ with $b^{\phi(g)}$; no correctness assumption about a producer's transversal is required. Orbits of $\phi(P_a)$ and $\phi(P_j)$ give the sets in (1). The formula constructs the base neighbour set, and the t_x transport it to every adjacency row. Every specified generator is checked on every row.

At all twenty-four anchors, the stabilizer of S_j in E is exactly $\phi(P_j)$ for every j. Thus the equivariant map from type-j objects to their shadows is injective: its fibres are the cosets of the shadow stabilizer modulo the object stabilizer. Consequently a typed automorphism fixing all anchor points fixes all objects. This proves faithfulness of restriction to each anchor. No refinement by additional types is needed for these fifteen applications.

Full graph groups and normality. A complete nauty calculation gives the order $q|G|$ in the table for each of the twenty-four graphs. The integer-order extraction is exact in the specified range, and every returned generator is checked against all adjacency rows. Let D be their literal permutation group. The verifier checks its order, $E \leq D$, and the two conditions

$$e^z \in E \quad (e \in \mathcal{E}, z \in \mathcal{D}), \qquad [z, w] \in E \quad (z, w \in \mathcal{D}),$$

where $\mathcal{E}$ and $\mathcal{D}$ are the displayed generating sets. The first condition implies $E \lhd D$: conjugation gives inclusion, hence equality by finiteness. The second makes D/E abelian. The acquired G is perfect, also checked in its original model, so $E = E' \leq D' \leq E$. Therefore $D' = E$ and E is characteristic in D.

A correlation carries each anchor graph to the graph of its image anchor, hence carries their derived groups to each other. If c is a correlation and $g \in G$, the restriction of cgc^{-1} to the image anchor equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes the anchor pointwise and is therefore trivial by faithfulness. Thus the specified G is normal in the full correlation group.

Completing the geometry groups. Put $B = \bigcap_j P_j$, and let S, T be the ordered and setwise stabilizers of the parabolic family in A. The acquired coset normalizer theorem now applies to the full geometry groups and gives

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The calculation considers every type permutation. Necessary order tests only discard impossible cases. For each remaining permutation, successive transporter calculations retain precisely the automorphisms carrying all parabolics already placed to their targets. Positive maps are checked

on the whole family. The resulting orders, in the order of the theorem, are

$$(|B|,|S|,|T|) = (660,660,1320),\ (1344,2688,2688),$$
$$(1536,3072,3072),\ (12,12,24),\ (12,12,12),\ (12,12,12),$$
$$(16,16,16),\ (256,512,512),\ (256,512,1024),$$
$$(4,4,8),\ (4,4,4),\ (4,4,8),\ (4,4,8),$$
$$(64,128,256),\ (3,3,6).$$

In every row $S \cap G = B$, so the typed group acts faithfully by conjugation on G and is identified with G or A by its order. The full centralizer of G is the actual quotient $(T \cap G)/B$. It has order two precisely for representatives 7,33,73 and order one otherwise. In those three cases the centralizer is central, meets the typed group trivially, and together with it has the full correlation-group order. This gives the asserted direct products. In every other row conjugation is faithful on the full correlation group and identifies it with G or A.

Attribution and verification boundary. The classification, source coordinates, shadow criterion and normalizer theorem are reused from the reviewed material. The contribution here is the fifteen explicit applications, the evaluation of pair counts using transitivity of a single shadow, and the use of a perfect subgroup with abelian quotient to verify the required characteristic subgroup. These are elementary deductions; no claim of historical novelty is made. GAP's exact group algorithms and nauty's complete upper bounds remain computational premises. Literal small coset models check (1) against direct incidence counts, including repeated shadows. Source pages 3,4,8,9,11,18,19,20,21 were inspected in the retained edition. Abstract parabolic and Borel names, other geometries and the whole manuscript remain outside this component. No conjecture resolution, new Lean verification or compression measurement is asserted.